



Feature Article

Amidase activity of phosphonate TSA-built polymer catalysts derived from organic monomers in the amidolysis of amino acid *p*-nitroanilides

Divya Mathew^a, Benny Thomas^{a,b}, K.S. Devaky^{a,*}^a School of Chemical Sciences, Mahatma Gandhi University, Kottayam, 686 560, India^b Department of Chemistry, St. Berchmans College, Changanacherry, 686 101, India

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ABSTRACT

Highly crosslinked transition state analogue imprinted macromatrix polymer catalysts having imidazole, carboxyl and hydroxyl functional groups in the catalytic sites were synthesized as chymotrypsin mimics using achiral organic monomers 4-vinylimidazole, methacrylic acid, allyl alcohol and phenyl-1-(N-benzyloxycarbonylamino)-2-(phenyl)ethyl phosphonate (transition state analogue of ester and amide hydrolytic reactions) as the template. The catalytic properties of the enzyme mimics were investigated in the amidolytic reactions of L-amino acid *p*-nitroanilides and correlated to the amidase activities of the catalysts derived from chiral methacryloyl-L-amino acid monomers methacryloyl-L-histidine, methacryloyl-L-aspartic acid and methacryloyl-L-serine. A two-fold enhancement in rate acceleration, substrate specificity, substrate shape-selectivity and stereoselectivity was observed for polymers made up of flexible amino acid monomers compared to the copolymers of organic monomers. The pre-polymerization complex of TSA with methacryloyl-L-amino acid monomers fabricates specific 3D-memory cavity preferentially of L-enantiomer of the TSA in the polymer matrix while the achiral organic monomers designs both L- and D- cavities. The effect of crosslink density on the catalytic efficiencies of the polymer catalysts was also investigated. Replacement of allyl alcohol by vinylpyridine afforded catalyst with better enzymatic activity.

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1. Introduction

Molecularly imprinted polymer catalysts (MIPs) have received considerable attention due to their thermal and pH stability and increased shelf life where natural enzymes may lose activity [1–4]. The proteolytic enzyme chymotrypsin continues to be a good model of MIP catalysts. Imidazole group of histidine residue is the essential catalytic group in the active site of the hydrolase protein chymotrypsin which cleaves the phenylalanine, tryptophan and tyrosine residues in peptide chains. Based on the theory of transition state stabilization for ester hydrolysis, phosphonic monoester TSA is generally used as a template molecule in molecular imprinting [6–12]. Even though there are many reports on MIP catalyzed esterolytic reactions, amide or peptide hydrolysis reactions using artificial enzyme mimics are less studied. We

have reported the amidolysis of phenylalanine *p*-nitroanilides using phosphonate TSA imprinted copolymers of amino acid monomers-methacryloyl-L-histidine, methacryloyl-L-aspartic acid and methacryloyl-L-serine [13]. Phosphonate analogue imprinted polymers with 4-vinylimidazole [14–16], methacrylic acid [9,17] and 4-vinylpyridine [18] have been used for esterolytic reactions. The present work describes the design and synthesis of catalytically active phenyl-1-(N-benzyloxycarbonylamino)-2-(phenyl)ethyl phosphonate (TSA). The individual and combined effects of 4-vinylimidazole, methacrylic acid, allyl alcohol and 4-vinylpyridine were studied in detail.

2. Experimental

2.1. Materials and methods

Dicyclohexylcarbodiimide (DCC), ethylene glycol dimethacrylate (EGDMA) and 4-vinylpyridine (VP) were purchased from Sigma Aldrich, USA. Z-L-Phenylalanine, methacrylic acid (MAA), allyl alcohol (AA), benzylcarbamate, triphenyl phosphite and phenyl-

* Corresponding author at: School of Chemical Sciences, Mahatma Gandhi University, Priyadarshini Hills, P.O Kottayam, 686 560, Kerala, India.

E-mail address: ksdevakyscs@yahoo.com (K.S. Devaky).